

Aliaksei Ivanou | Senior C++ Software Engineer

Qt · Embedded Linux · HMI · Yocto · Networking

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Relocating to Poland | Seeking UoP roles in Warsaw and across Poland

Polish D visa: 26 Sep 2026 - 25 Sep 2027 | UoP via employer-registered oświadczenie

Senior C++ Software Engineer with 6+ years of commercial C/C++ experience and 13+ years across software engineering and R&D. Specialized in Qt-based HMI, Embedded Linux, Yocto/BitBake, NXP i.MX, and SIP/RTP/WebRTC real-time communication. Hands-on experience designing production UI architecture and delivering device integration, networking, Linux debugging, and automated testing, with code reviews, technical mentoring, onboarding, and technical interview support in a 6-person engineering team.

Technical Skills

- **C++:** C++11/14/17, STL, C, OOP, RAII, smart pointers, object lifetime management
- **Architecture & Engineering Practices:** design patterns, MVC/MVVM-style UI architecture, code reviews
- **Linux:** Embedded Linux, Yocto/BitBake, NXP i.MX, systemd, Linux troubleshooting, service configuration, shell/Bash
- **Qt/HMI:** Qt 5.15, Qt Widgets, Qt Model/View, Qt signals/slots, QObject, QSS
- **Networking & Real-Time Media:** TCP/IP, SIP, RTP/SDP, WebRTC, WebSockets, liblinphone, mediasoup, GStreamer, V4L2, SNMP, DHCP, TFTP
- **Build, Debugging & CI/CD:** CMake, GCC, gdb, Valgrind, core dump analysis, Git, GitLab CI, Jenkins, SSH
- **Testing:** Google Test, QTest, unit/integration testing, TDD
- **Databases:** PostgreSQL, SQLite, MySQL
- **Other:** C#/.NET 8, JavaScript, Java/Scala

Experience

C++ Software Engineer | [Innowise Group](#)

12/2024 - Present

Global software engineering and IT services company.

- Refactored a legacy Qt Widgets SIP call-flow toward an MVC/MVVM-style UI architecture, separating UI, call state, user actions, media negotiation, and Linphone-SDK integration while preserving Qt signal/slot compatibility.
- Extended and customized Linphone-SDK (liblinphone) backend logic to implement SIP failover and backup-server connectivity for improved device reliability.
- Developed and hardened SIP/WebRTC calling and video-conferencing flows, including multi-call handling, remote hangups, conference management, media switching, participant management, and audio/video controls.
- Implemented SNMP/DHCP/TFTP/SQLite device-provisioning workflows while migrating a WinForms utility from .NET Framework 4.8 to .NET 8.
- Integrated and deployed media services on Embedded Linux, including mediasoup/systemd configuration, Yocto/BitBake customization, NXP i.MX integration, and device-side service configuration.
- Added QTest unit/integration coverage for SIP, media, and networking flows, including offscreen execution on the embedded target.
- Performed code reviews, contributed to task planning, and provided onboarding, technical mentoring, and interview support as the engineering team grew to 6 people.

Tech Stack: C++17, Qt 5.15.7/Qt Widgets, Embedded Linux, Linphone-SDK, SIP/WebRTC, mediasoup, Yocto/BitBake, NXP i.MX, systemd, QTest, C#/.NET 8, CMake, GCC/gdb.

Global digital engineering and IT services company.

- Developed and maintained C/C++ modules across two production enterprise systems: Energy ERP (C++17) and Digital Library (C89), and performed peer code reviews in distributed international engineering teams.
- Diagnosed production and integration issues on Linux systems hosted on AWS EC2 and Azure VMs using SSH, gdb, and core dump analysis.
- Digital Library: Maintained legacy C89 modules and resolved integration defects between the C core and Java/Scala services.
- Energy ERP: Optimized PostgreSQL schema and SQL queries, reducing search response time from 1.5s to 300ms.
- Maintained JavaScript integration modules connecting backend data flows with the Web UI.

Tech Stack: C89, C++14/17, PostgreSQL, Java 8/11, Scala, JavaScript, Linux, gdb, AWS/Azure, CMake, Jenkins.

C++ Software Engineer | RIFTEK**03/2020 - 05/2020**

Industrial metrology and 3D scanning software.

- Developed a 3D reconstruction R&D prototype using C++17, Qt, and Point Cloud Library (PCL).

Tech Stack: C++17, Qt, PCL.

MATLAB Algorithm Developer | PELENG**08/2012 - 01/2020**

Aerospace, Earth-observation and image-processing R&D.

- Developed Earth-observation and UAV image-processing algorithms, including feature-based image stitching, stabilization, debayering, and sensor fusion.
- Implemented custom lossy satellite-image compression, achieving a 3-4x compression ratio.
- Developed image-type prediction and quantization algorithms, improving image-quality metrics by ~30% and estimated Earth-observation accuracy by 15-20%.

Tech Stack: MATLAB, Git, MS Access.

Education

- BSc in Radiophysics - Satellite information systems and technologies, [Belarusian State University, Faculty of Radiophysics and Computer Technologies](#) | 2007 - 2012

Languages

English B2 · Polish B1 · Swedish A2 · Russian Native · Belarusian Native